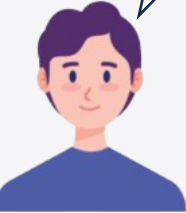


Search recent literature on KRAS G12C inhibitor resistance in NSCLC.



User

Orchestrate Agent
(Planning & Dispatch)

Planning

1. Search literature for mechanisms.
2. Extract key findings.
3. Structure into categories.
4. Include citations.
5. Deliver summary.

Dispatch

Task:

- ❑ Search recent peer-reviewed literature on KRAS G12C inhibitor resistance mechanisms in NSCLC.

Retrieve Agent
(Initial Search)

Tool Call

Query

KRAS
G12C
inhibitor
resistance
NSCLC



Tool Response

Initial Results



List of papers,
abstracts.

Orchestrate Agent
(Review & Refine)

Review



Initial results are
too broad, need
specific PMIDs
and details.

Refinement
Request

Task:

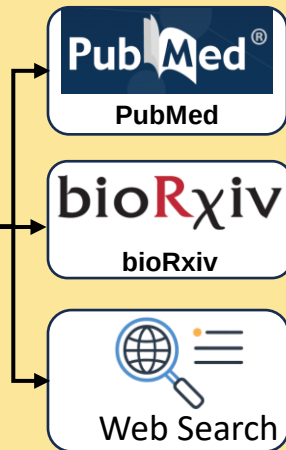
- ❑ Refine search for specific PMIDs/DOIs and detailed resistance mechanisms from landmark studies.

Retrieve Agent
(Refined Search)

Tool Call

Query

PMID
33431478
KRAS
G12C



Tool Response

Detailed Results



Specific PMIDs,
verified mechanisms
(Y96D, R68S, etc.),
off-target pathways.

Orchestrate Agent
(Final Analysis & Output)

Review



Information is
sufficient and
structured

Output



**KRAS G12C Inhibitor
Resistance in NSCLC**

1. Key Resistance Mechanisms
A. On-Target KRAS Mutations
Secondary mutations in KRAS directly interfere with inhibitor binding:
Y96D/C: Disrupts hydrophobic interactions in the switch-II pocket, reducing sotorasib/adagrasib affinity.
R68S/L: Impairs covalent bond formation with the inhibitor.
H95D/Q/R: Alters nucleotide-binding dynamics, promoting GTP-bound state persistence.
Source: Canon et al., Cancer Discovery (2021); PMID [33431478](#)
Source: Lanman et al., Nature (2021); PMID [33431479](#)
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